Group 3 - Space Complexity

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Method Selection and Planning

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Software Engineering Methods

Programming Methodology

When working on a large project it is essential to use an appropriate programming methodology to keep the project organised and delivered on time. We researched 3 different programming methodologies and compared their strengths and weaknesses.

We first looked into the waterfall methodology, this is where you follow a linear development path going from plan to final product. It is divided into different phases of development such as Requirements and Implementation which needs to be reviewed and signed off before moving onto the next phase. Although this is the simplest to understand and use, it doesn't promote change because once you have signed off on a phase you can't go back and change it. For this reason, we didn't choose to use the waterfall methodology as we will constantly be suggesting changes and new ideas to improve the project.

Following this, we researched the spiral methodology. In this methodology you repeatedly work through development cycles which consist of 4 stages; Planning, Risk analysis, Engineering and Evaluation. This allows for constant iteration and improvement of the software since flaws and new ideas are always being evaluated and implemented. However, this methodology requires a lot of planning and management which is intended for larger projects alongside a focus on risk analysis which isn't needed for this project.

Finally, we settled on the agile methodology since it best matched the needs of our project. You initially come up with a plan and a list of requirements to outline the scope of the project. Then you create a backlog of tasks which need to be completed in each week's sprint. A sprint consists of 5 stages; Design, Development, Testing, Deployment and Review. This allows us to have greater flexibility during the development process and allows our team to collaborate effectively on the project due to its need for constant communication between team members.

Collaboration and Development Tools

As a team, we decided to go with the Google Suite as our main set of tools due to their great collaboration features, ease of use, and native version control features. We explored other sets of tools such as the 'Microsoft 365' set of tools, however, the university already provides us with the Google Suite and it would be too much of a hassle to migrate over. Following this, we decided that our deliverables would be written in Google Docs. In Assessment 1, the original group used Google Slides for their logbook and Google Sheets to track their work distribution, whilst in Assessment 2 we used Google Docs to keep track of the meeting minutes which detailed everyone's progress and tasks. We also used Google Drive as it helped to gather all of our deliverables in a shared space, allowing for a convenient and collaborative environment where team members could work on documents in real-time with version control tools.

For implementation, we decided to use Github as our code version control system. It allowed for the code to be stored in a shared location and had tools to facilitate collaboration between team members as well as useful CI features. This ensured that our code was always up to date between members working on the code.

We decided to use Visual Studio Code as we had members on the Implementation team fairly new to programming. Visual Studio Code is great for novice users and our members had prior experience with it due to our previous software modules.

In addition to this, we also used IntelliJ IDE to develop our code. We had members with different preferences and believed it'd be more efficient for them to work with tools they were already familiar with. IntelliJ IDE features code completion, refactoring and debugging tools. This greatly improved our implementation productivity.

To create our structural, behavioural and Gantt chart diagrams we went with a mix of using PlantUML with the Visual Studio Code diagram generation extension and Goodnotes 6. For Assessment 2, we worked off of the original PlantUML script provided by the original group. Using PlantUML achieved a clear view of the different tasks and their timeframes within each deliverable, this helped support our time management progress. It also tracked the dependencies between each task throughout the project lifecycle. Aside from this, we used Canva to create a work breakdown structure to identify the different tasks within each deliverable. This gave us a good overview and understanding of the task priority sequences and dependencies.

For the artwork, we used a mixture of Aseprite and Photoshop to create the UI and sprites owing to their versatility and plethora of features. To maintain a seamless and consistent display of art style from the original sprites, we utilised DALL-E to create the new features following Assessment 2. Using generative AI, we fed it screenshot examples of the original sprites and gave it detailed descriptions of what results we wanted to achieve, specifying that we wanted it to follow the same art style input.

As for communication channels, we chose WhatsApp (original group) and Discord (Assessment 2 group) to be our main points of contact. Discord allowed for the transferring of larger files and the ability to be able to share your screen during team calls, whilst WhatsApp was already installed on our members' phones which meant they would be quick to respond to any important messages if needed, which is something very important within the agile methodology; response and engagement.

Team Organisation

Roles and Responsibilities

When it came to team organisation we initially wrote down what each team member wanted to work on in the project and created a spreadsheet to equally distribute the work. From this we created an initial Gantt chart, outlining when different parts of the project were to be started and finished. We also created a mini Gantt chart each week which showed the current week's tasks alongside an idea for what was going to get done next week (All weekly Gantt charts are available on the website under the planning tab).

	Website	Requirements	Architecture	Method Selection and Plannning	Risk Assessment	Implementation	Total
James	0	0	5.5	0	5	5	16
Sharlotte	0	6.666666667	5.5	0	0	0	12
Tommy	1	6.666666667	0	3.333333333	0	5	16
Damian	1	0	5.5	3.333333333	0	5	15
Elliott	0	6.666666667	0	0	5	0	12
Ren	1	0	5.5	0	0	5	12
Zhenggao	0	0	0	3.333333333	0	5	8

	Marks	Members Allocated
Website	3	3
Requirements	20	3
Architecture	22	4
Method Selection and Planning	10	3
Risk Assemssment	10	2
Implementation	25	5
Peer Assessment	10	7
Total	100	

Each week we had 2 meetings, one on a Monday and the other on a Friday. On Mondays, we discussed how the project was going and outlined the tasks for the week alongside creating that week's Gantt chart. On Fridays, we reviewed that week's work and suggested any changes which could be made over the weekend before the next meeting. In each meeting, we updated our logbook which kept track of attendance and what was discussed in the meeting.

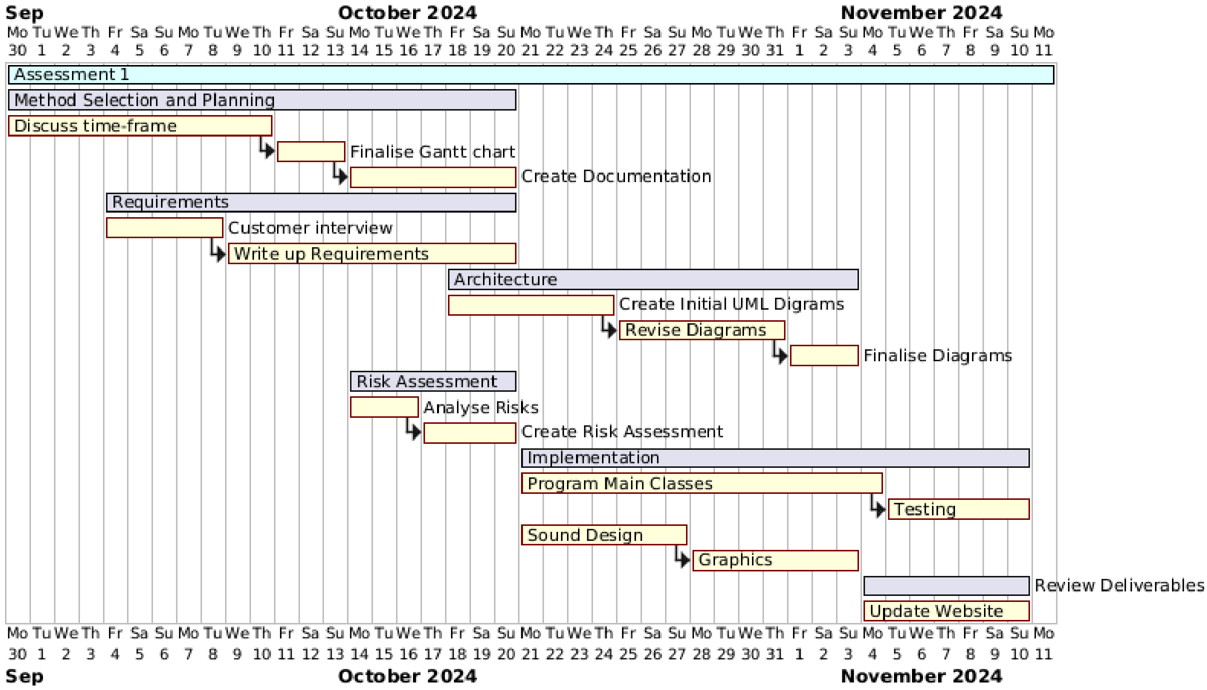
From the beginning of Assessment 2, we discussed and negotiated our respective roles and responsibilities. Following on from our previous roles in Assessment 1, we concluded that we would stick to the same roles in Assessment 2 with a few minor changes. This was because we felt comfortable and more knowledgeable in our respective areas, allowing us to work more efficiently and effectively. Due to the mark contribution for different sections, we also negotiated who should take up the newly added deliverables to ensure everyone had on average similar mark contributions.

Throughout the allocation of new roles, we made sure to consider the low bus factor through the number of people working on the same deliverable. This was to support better risk mitigation across the project lifecycle. An example of this is our Implementation and Software Testing teams, whereby we decided on keeping the same three people for the two deliverables as these sections weighed heavier than most and that the marks would be better distributed this way. This also allowed for a more seamless and consistent style and quality as the two sections are closely intertwined, with iterative checks between them.

Each deliverable had its respective single leader to oversee and manage its tasks. Doing so supports a clearer vision/overview of the deliverable and reduces any confusion about each member's work.

Project Plan–Assessment 1

Overview



When planning the project we created an initial Gantt chart, planning when work would be completed.

Week 1 23/9/24

During week 1 we are planning on getting to know each other and each other's strengths and weaknesses. We are also hoping to talk about what tools we are going to use and how we are going to tackle the product brief as well as how we are going to share the workload.

Week 2 30/9/24

During week 2 we are planning on having set up all of our collaboration tools that we are going to use throughout the project and start discussing who should focus on which tasks. During this week we are also planning on brainstorming questions for our client meeting which will occur on the 8th of October.

Week 3 7/10/24

During week 3, we are planning on writing up our requirements, depending on how the client meeting goes, hopefully these should be done by the end of the week as this would allow us to slowly start thinking about architecture and implementation. We are also hoping to finalise the weekly plan gantt charts by this point. Somewhere around this time we are also planning on putting up the website so we can track our weekly progress.

Week 4 14/10/24

During week 4 we are planning on creating our risk assessment as well as starting work on architecture. Hopefully if we can get these things done early we will be able to start implementation a bit earlier than first hoped..

Week 5 21/10/24

During week 5 we are hoping to be well into programming the game as this would mean we would have enough time if anything were to go wrong, or take longer than expected. We are also hoping to come to the end of our architecture document, as that would mean there would be nothing else to do but develop the game. Around this time we are also planning to start developing the assets for our game.

Consolidation Week 28/10/24

By the end of consolidation week we want to be fully finished with all of the documentation and focus on finishing creating assets for the game and the code itself. If all of this is done we will start focusing on making sure the website is up to date.

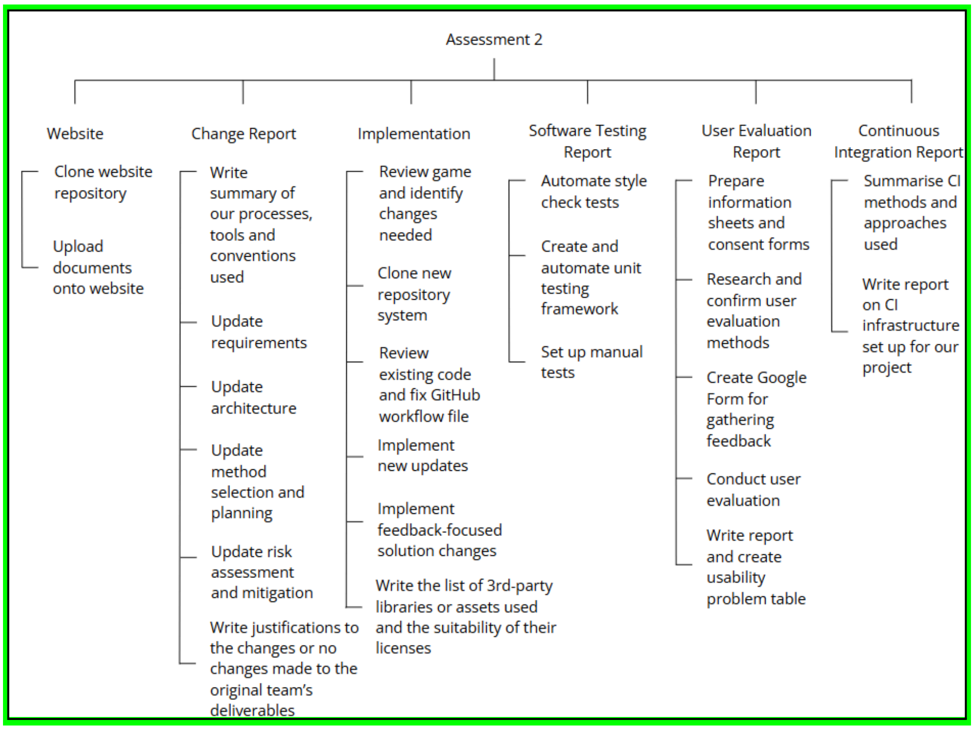
Week 6 4/11/24

By week 6 we are hoping to have finished everything there is to do, and will be making sure that the formatting of each deliverable is as it should be ready for submission, as well as updating the website to contain all of the relevant information in chronological order.

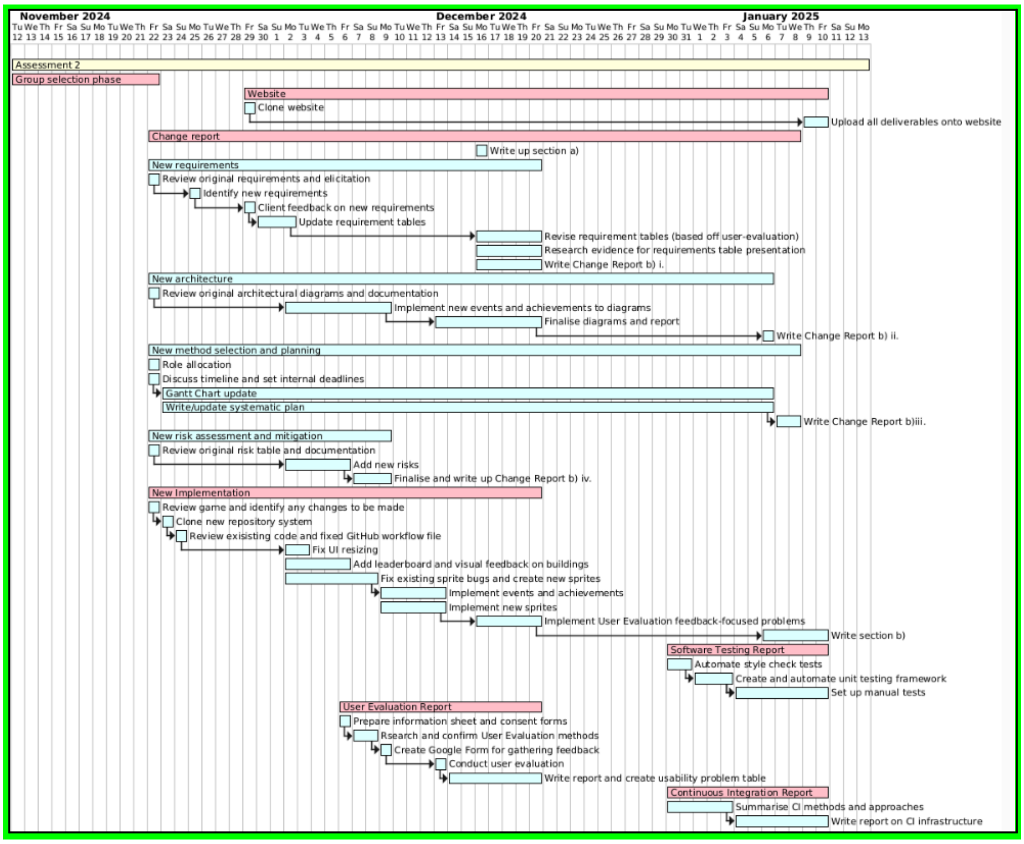
Project Plan–Assessment 2

Overview

Assessment 2 work breakdown structure



Assessment 2 Project Plan (Gantt chart)



Week 8: 18th - 22nd Nov

This week was the group selection phase. However, we did begin reviewing every deliverable and the game in more detail during the Friday practical session. This was to help us gain a brief overview of what needs to be changed (or not). In addition, we sorted out the management aspect and appointed our project manager to keep track of tasks and the general schedule. We also discussed the timeline and decided on our internal deadlines to ensure we deliver our assessment on time.

Week 9: 25th - 29th Nov

After asking our client and getting feedback from them, we were able to finalise what new requirements we needed to add and which original ones needed changing. E.g. We asked about the required events and what they would like to see, by doing so, we could use it to back up our events to be implemented.

As requirements were firmed, we could begin our implementation and update the requirements table and architectural diagrams. We cloned the new website, reviewed the existing code, and fixed the GitHub workflow file.

To keep things organised, we kept a document of our timeline and updated our systematic plan alongside the Gantt chart. These two will be updated regularly up until our internal deadline of 10/01/2025. We also adjusted the deadline for the change report as it is dependent on the Method section, and in general, cannot be finalised until all deliverables from assessment 1 have been updated.

Week 10: 2nd - 6th Dec

We discussed how to identify and manage the changes we make to the deliverables, by which we decided to highlight the things we changed and added in different colours.
We continued to implement new features like events and achievements to our UML architectural graphs and updated the change report with respective edits. We analysed and formalised the new risks and updated the change report accordingly. For implementation, we worked on adding new features to the game.

Week 11: 9th - 13th Dec

We discussed and planned our user evaluation, including preparing documents and creating a Google Form. By Friday (our last practical session), we successfully implemented the prototype version of our chosen game, following the Assessment 2 product brief. This allowed us to begin our user evaluation phase, whereby we had 7 volunteers play the game and answer our short Google Form. The deadline for architecture is set to end alongside finishing the final version of the game with feedback-focused updates, to ensure that the architecture reflects and represents the implementation fully.

Week 12: 16th - 20th Dec

From the open-ended feedback gathered by our Google Form, we were able to pinpoint specific areas that needed fixing or extra details to add to allow for a better user experience. The implementation and architecture sections were finalised simultaneously this week.

For Requirements, we revised the requirements table to include the new user-feedback requirements and added extra research evidence for its presentation. This enabled us to wrap up the change report for its respective section.

Our user feedback was compiled and analysed to create a usability problem table with scales to measure and prioritise the problems we collected; the user evaluation report was completed in the set timeframe.

Section 2a of the change report was written up this week.

Week 13: 23rd - 27th Dec

As this was Christmas week, most of the team was busy due to personal commitments, so little was achieved. Hence, there is no update on the Gantt chart.

Week 14: 30th Dec - 3rd Jan

Our team's progress slowed down significantly due to a lack of communication and slow responses from members as it was still the holidays. This meant that we started on the latter sections of Assessment 2 later than we'd hoped.

Throughout the implementation process, our team consistently practised continuous integration, whereby our members updated and shared their progress almost daily and multiple times a day. This week, we started writing the first part of the Continuous Integration report.

We started on software testing, where we added style check testing and automated it. Afterwards, we manually went through the entire project refactoring it to pass all the style tests. The framework for unit testing was created and also automated during this week. We had three members working on software testing as a part of the implementation section, hence there was a handover, resulting in a delay in the workflow for this area.

Week 15: 6th - 10th Jan

This week marks the final week of our Assessment 2 timeline as we have set the 11th Jan as our internal deadline for our submission. This also means that the Method selection and planning section will be finalised this week (as the Gantt chart will be done), along with its respective change report section.

We are still working on Software Testing, writing the report and continuing to set up unit and manual tests along with its test plans.

To finish our Continuous Integration section, we are writing up the CI infrastructure we have set up for our project.

We will upload everything onto the website after finalising all deliverables (such as the remaining Implementation part b and Architecture change report).

Week 16: 13th Jan

We set our internal deadlines at the beginning of Assessment 2 to support a timely submission. We managed to submit our project two days before the external deadline. There is no update to the Gantt chart as we have now finished Assessment 2.

Weekly snapshots of how our project plan evolved are uploaded on our website: https://vikingz-updated.vercel.app/pdfs/timeline.pdf